

Wazuh SOC Detection Lab

Windows Endpoint Detection Engineering — Attack Simulation & Custom Rule Validation

This report documents a hands-on Security Operations Centre (SOC) detection lab built using Wazuh SIEM. Four Windows-based attack techniques were simulated on a live endpoint and detected using custom Wazuh rules. The objective is to demonstrate practical skills in attack simulation, log analysis, and behavior-based detection engineering.

Platform	Scenarios	Custom Rules	Framework
Wazuh SIEM + Windows 11	4 Attack Techniques	2 Detection Rules	MITRE ATT&CK

1. Overview

This lab replicates a real-world SOC analyst workflow by executing four common Windows-based attack techniques on a monitored endpoint and validating that Wazuh generates accurate, actionable alerts. The core objective is to demonstrate the transition from passive event collection to **behavior-based, pattern-driven detection**.

Lab Environment

SIEM Manager	Wazuh (latest) on Ubuntu Linux — IP: 172.28.2.49
Monitored Endpoint	Windows 10 Desktop — hostname: WIN10-LAB (Agent ID: 001)
Agent Mode	Real-time log forwarding via Wazuh agent
Log Source	Windows Security Event Log
Rule Language	Wazuh XML (custom rules in /var/ossec/etc/rules/local_rules.xml)
Framework	MITRE ATT&CK Enterprise v14

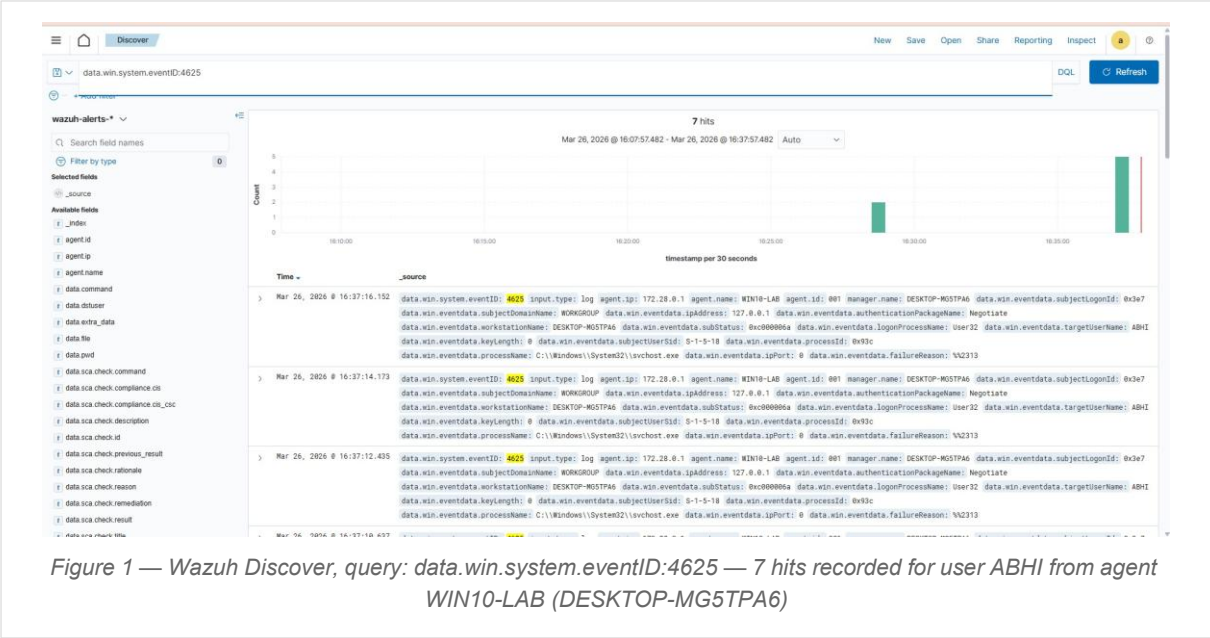
2. Attack Scenarios

Four attack phases were executed sequentially on the Windows endpoint. Each attack is mapped to a Windows Security Event ID and a MITRE ATT&CK technique.

2.1 Failed Logon Attempts — Event ID 4625

Attack Type	Password guessing / brute-force simulation
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Method	Multiple incorrect password entries for local account ABHI
MITRE	T1110 — Brute Force
Result	7 failed logon events captured in Wazuh Discover within a 30-minute window

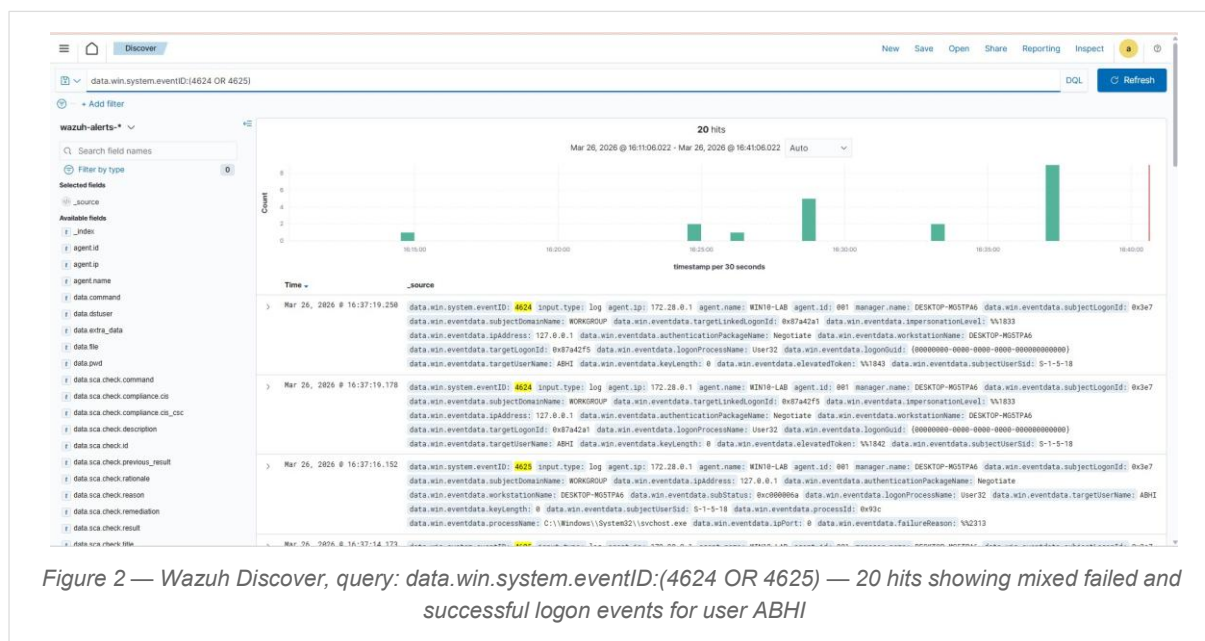


SOC Analyst Note

Windows generates a 4625 event for every failed logon, including benign mistyping. A single-event alert produces excessive noise. The real detection value comes from counting the frequency of failures within a short timeframe, the basis for Rule 100001.

2.2 Successful Login After Failures — Event IDs 4624 + 4625

Attack Type	Account compromise following credential guessing
Method	Multiple failed logons followed by a correct password entry
MITRE	T1078 — Valid Accounts
Result	20 combined hits (failures + success) captured; foundation for correlation detection

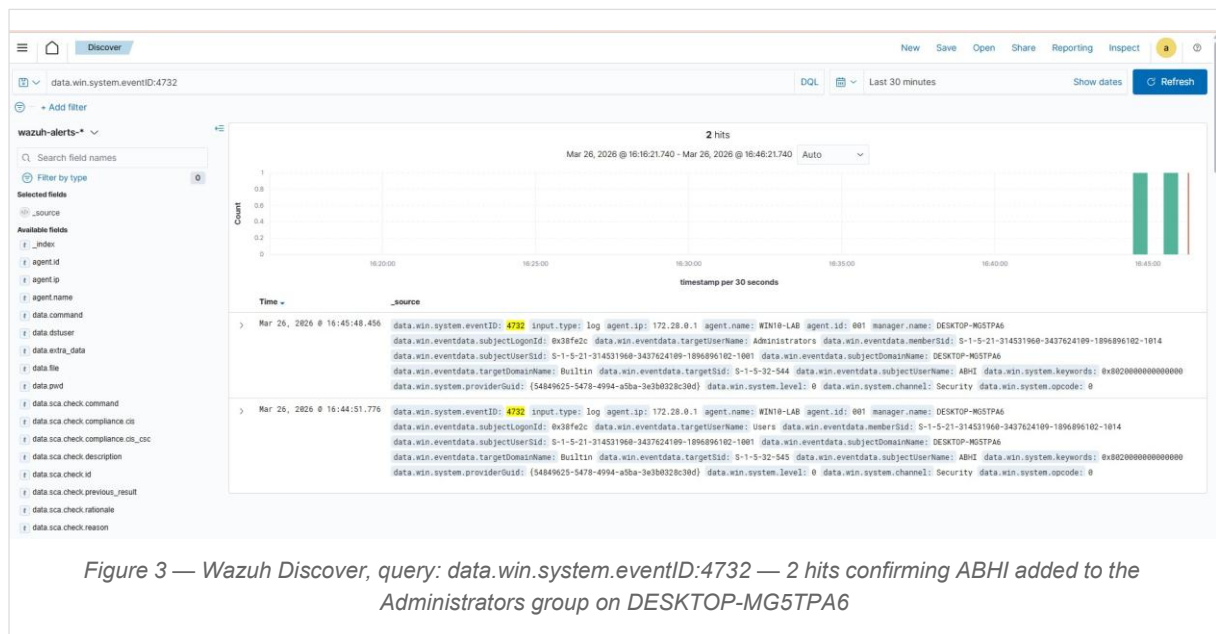


SOC Analyst Note

Failures followed by a success is the canonical account compromise pattern. Detecting it requires correlating two separate event types. A future composite rule chaining Rule 100001 (failures) and Rule 100002 (success) will complete this detection chain.

2.3 Privilege Escalation — Event ID 4732

Attack Type	Local privilege escalation via group membership manipulation
Command Used	<code>net localgroup Administrators testuser /add</code>
MITRE	T1098 — Account Manipulation
Result	Wazuh detected group change: user ABHI added to Administrators (Builtin)



SOC Analyst Note

Adding an account to a privileged group is one of the highest-confidence Windows security events with near-zero false-positive rate in a standard environment. Event 4732 should always trigger an immediate high-severity alert.

2.4 Audit Log Clearing — Event ID 1102

Attack Type	Defence evasion — destruction of forensic evidence
Command Used	<code>wevtutil cl Security</code>
MITRE	T1070.001 — Clear Windows Event Logs
Result	Wazuh captured event 1102 before logs were erased — evasion attempt detected

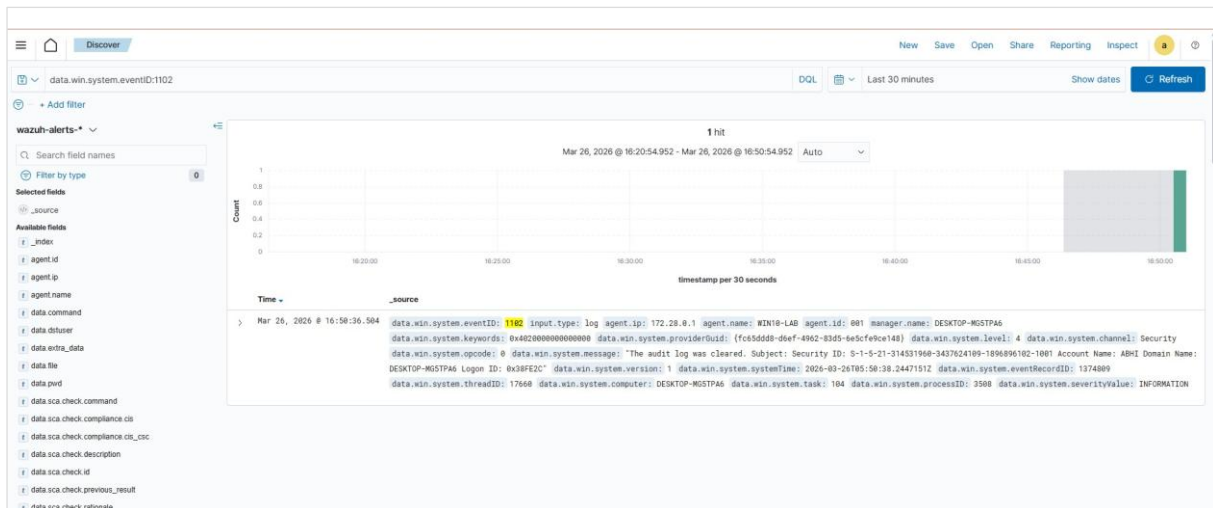


Figure 4 — Wazuh Discover, query: `data.win.system.eventID:1102` — 1 hit confirming audit log cleared by account ABHI at 16:50:36

Why the evasion failed

The Wazuh agent forwarded event 1102 to the SIEM before the Windows log was erased. Clearing local logs cannot remove data already in the SIEM. This demonstrates the importance of centralised, log forwarding as a fundamental control against log-tampering attacks.

3. Detection Engineering

Two custom Wazuh detection rules were authored, debugged, and validated. Both rules reside in `/var/ossec/etc/rules/local_rules.xml` on the Ubuntu Wazuh Manager.

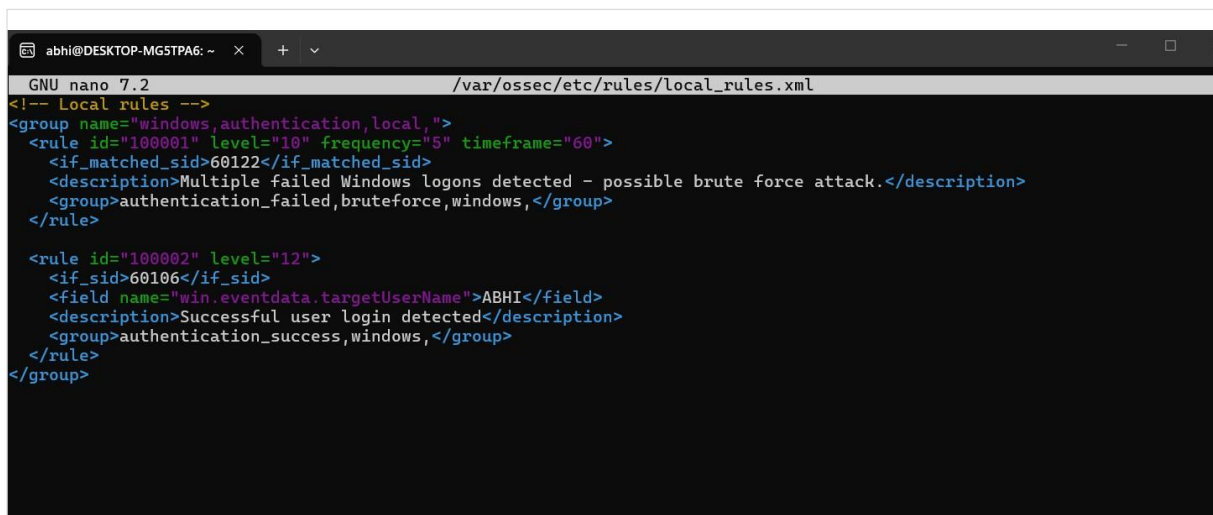


Figure 5 — The final `local_rules.xml` file open in GNU nano on the Wazuh Manager, showing both custom rules (100001 and 100002)

3.1 Rule 100001 — Brute Force Detection

Triggers when five or more failed Windows logon events occur within a 60-second window. This moves detection from reacting to individual wrong passwords to identifying sustained credential-guessing behavior.

```
<group name="windows,authentication,local,">

<rule id="100001" level="10" frequency="5" timeframe="60">
<if_matched_sid>60122</if_matched_sid>
<description>Multiple failed Windows logons detected -
possible brute force attack.</description>
<group>authentication_failed,bruteforce,windows,</group>
</rule>

</group>
```

Attribute	Value
Rule ID	100001
Severity Level	10 — High
Trigger Logic	5 or more failures matching SID 60122 within a 60-second window
Detection Type	Behavior-based pattern detection (not single-event)
Alert Groups	authentication_failed, bruteforce, windows



Figure 6 — Wazuh Discover, query: rule.id:100001 — custom brute force rule triggered (1 hit), confirming behaviour-based detection is operational

3.2 Rule 100002 — Successful Login Detection

Captures real interactive user logins by matching SID 60106 (Event ID 4624), filtering out SYSTEM and background logon events to isolate genuine user sessions. Designed as the second leg of a brute-force-to-compromise correlation chain.

```
<group name="windows,authentication,local,">

<rule id="100002" level="12">
<if_sid>60106</if_sid>
<description>Successful Windows logon detected.</description>
<group>authentication_success,windows,</group>
</rule>

</group>
```

Attribute	Value
Rule ID	100002
Severity Level	12 — Critical
Trigger Logic	Successful logon matching SID 60106 (Event 4624)
Detection Type	Contextual — real user login vs. SYSTEM/background noise
Confirmed Fields	targetUserName: ABHI · logonProcessName: User32 · IP Address: 127.0.0.1

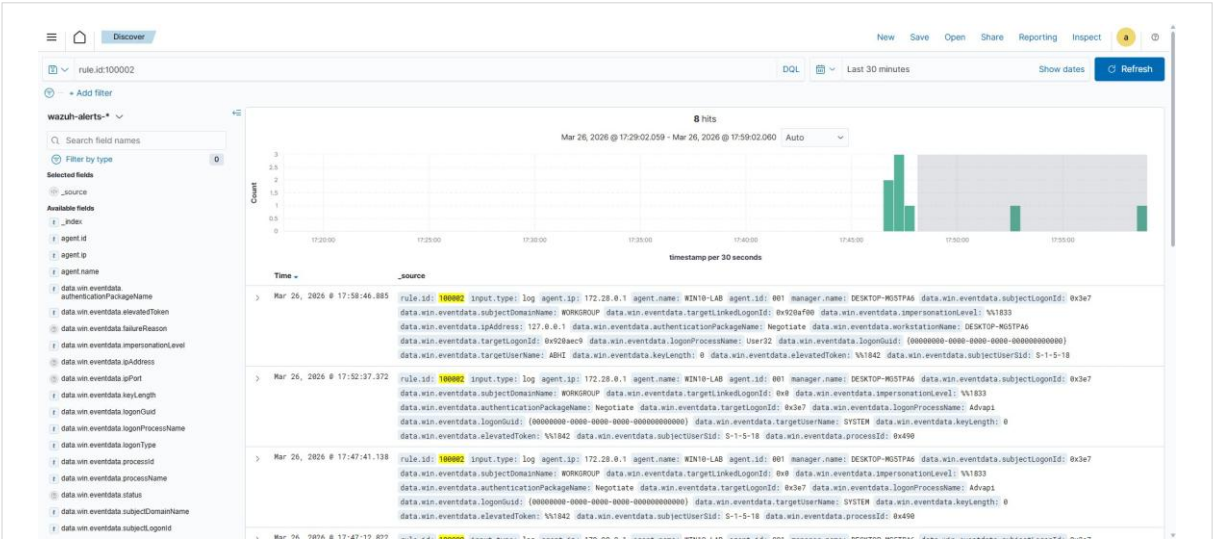


Figure 7 — Wazuh Discover, query: rule.id:100002 — 8 hits confirming successful login detection including real user ABHI (User32 logon process)

3.3 Rule Authoring Process and Debugging

Writing Wazuh XML rules requires iterative syntax validation before restarting the manager. The screenshot below shows the debug process, including real CRITICAL errors encountered (missing if_matched on frequency rule, invalid root element) and the steps taken to resolve them.


```

abhi@DESKTOP-MG5TPA6:~$ sudo systemctl start wazuh-manager
[sudo] password for abhi:

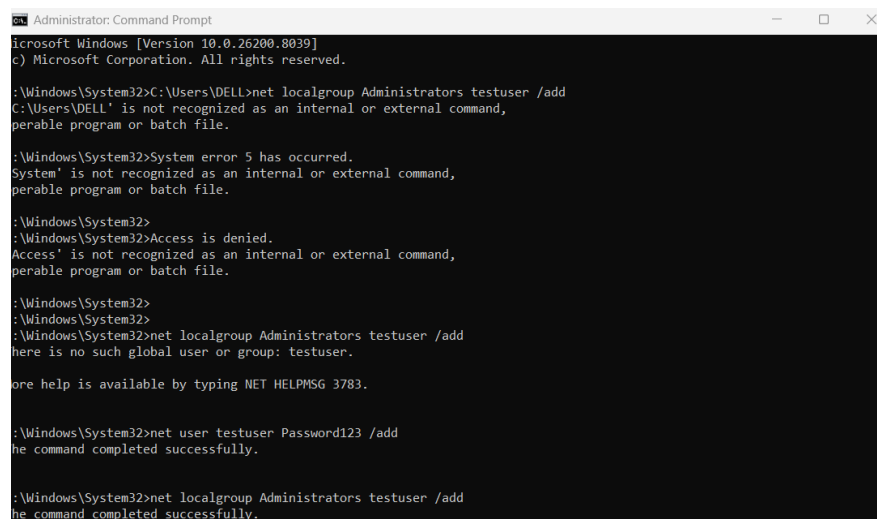
abhi@DESKTOP-MG5TPA6:~$ sudo systemctl start wazuh-manager
abhi@DESKTOP-MG5TPA6:~$ sudo systemctl start wazuh-dashboard
abhi@DESKTOP-MG5TPA6:~$ hostname -I
172.28.2.49
abhi@DESKTOP-MG5TPA6:~$ sudo nano /var/ossec/etc/rules/local_rules.xml
[sudo] password for abhi:
abhi@DESKTOP-MG5TPA6:~$ sudo nano /var/ossec/etc/rules/local_rules.xml
abhi@DESKTOP-MG5TPA6:~$ sudo systemctl restart wazuh-manager
Job for wazuh-manager.service failed because the control process exited with error code.
See "systemctl status wazuh-manager.service" and "journalctl -xeu wazuh-manager.service" for details.
abhi@DESKTOP-MG5TPA6:~$ sudo /var/ossec/bin/wazuh-analysisd -t
2026/03/26 17:00:23 wazuh-analysisd: ERROR: Invalid use of frequency/context options. Missing if_matched on rule '100001'
2026/03/26 17:00:23 wazuh-analysisd: CRITICAL: (1220): Error loading the rules: 'etc/rules/local_rules.xml'.
abhi@DESKTOP-MG5TPA6:~$ sudo nano /var/ossec/etc/rules/local_rules.xml
abhi@DESKTOP-MG5TPA6:~$ sudo /var/ossec/bin/wazuh-analysisd -t
abhi@DESKTOP-MG5TPA6:~$ sudo systemctl restart wazuh-manager
abhi@DESKTOP-MG5TPA6:~$ ^C
abhi@DESKTOP-MG5TPA6:~$ sudo nano /var/ossec/etc/rules/local_rules.xml
[sudo] password for abhi:
abhi@DESKTOP-MG5TPA6:~$ sudo /var/ossec/bin/wazuh-analysisd -t
2026/03/26 17:39:27 wazuh-analysisd: ERROR: rules_op: Invalid root element "rule".Only "group" is allowed
2026/03/26 17:39:27 wazuh-analysisd: CRITICAL: (1220): Error loading the rules: 'etc/rules/local_rules.xml'.
abhi@DESKTOP-MG5TPA6:~$ sudo nano /var/ossec/etc/rules/local_rules.xml
abhi@DESKTOP-MG5TPA6:~$ sudo /var/ossec/bin/wazuh-analysisd -t
abhi@DESKTOP-MG5TPA6:~$ sudo systemctl restart wazuh-manager

```

Figure 8 — Ubuntu terminal showing Wazuh manager startup, rule validation via `wazuh-analysisd -t`, CRITICAL syntax errors, and successful restart

4. Test Environment Setup

To trigger Event ID 4732 (privilege escalation), a test user account was created via Windows Command Prompt with Administrator privileges and added to the local Administrators group.



```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>net localgroup Administrators testuser /add
C:\Windows\System32>net user testuser Password123 /add
The command completed successfully.

C:\Windows\System32>net localgroup Administrators testuser /add
The command completed successfully.

```

Figure 9 — Windows Administrator CMD — creating testuser with net user, then adding to the Administrators group with net localgroup

```

# Step 1 - Create the test user account
net user testuser Password123 /add

# Step 2 - Add to local Administrators group (triggers Event ID 4732)
net localgroup Administrators testuser /add

```


5. MITRE ATT&CK Mapping

Technique ID	Name	Lab Scenario	Event ID
T1110	Brute Force	Multiple failed logon attempts	4625
T1078	Valid Accounts	Successful login after failures	4624
T1098	Account Manipulation	Admin group membership change	4732
T1070.001	Clear Windows Event Logs	Security audit log cleared	1102

6. Key Findings

1. Behaviour-based detection over single events

Windows generates thousands of authentication events per day. Single-event rules create noise, not signal. The shift to frequency-plus-timeframe correlation in Rule 100001: 5 failures within 60 seconds is the critical step toward SOC-grade detection quality. This is real detection engineering, not just log forwarding.

2. Correlation chain is partially built

With Rule 100001 (brute force failures) and Rule 100002 (successful login) both operational and validated, the lab is one rule away from a complete brute-force-to-compromise detection chain. This is the most common credential-based attack pattern observed in real-world incidents.

3. High-confidence indicators operate reliably

Privilege escalation (4732) and audit log clearing (1102) are attacker behaviors with near-zero false-positive rates in a standard enterprise environment. Both were captured cleanly in every test run, confirming Wazuh's detection pipeline is correctly configured.

4. Centralised log forwarding defeats evasion

Log clearing (1102) was detected because the Wazuh agent had already forwarded the event to the centralised SIEM before the local Windows log was erased. This demonstrates a fundamental architectural principle: attackers cannot retroactively remove data from a well-configured SIEM by clearing local logs.

7. Conclusion

This lab demonstrates practical SOC analyst capability across three domains: **attack simulation**, replicating real threat-actor techniques on a live Windows host; **log analysis**, querying and interpreting Windows Security events in Wazuh Discover; and **detection engineering**, authoring, validating, and debugging custom XML rules that detect attack patterns rather than individual events.

The most significant outcome is the transition from passive log collection to **proactive, pattern-based detection**, the same philosophy that underpins mature, real-world SOC operations. Every scenario simulated corresponds to a documented MITRE ATT&CK technique; every alert generated is evidence of a functioning detection pipeline. The lab is intentionally designed to be extended, making it a practical foundation for more advanced correlation rules and integrations.
